

Bill Of Materials (BOM):

Item	Qty	Unit Price	Total	Notes/Links
Cytron MOTION 2350 Pro (RP2350)	1	\$24.57	\$24.57	Digi-Key #3614-MOTION-2350-PRO-ND
Raspberry Pi 5, 4 GB	1	~\$130.00	~\$130.00	Adafruit product 5812; Farnell £82.09
TI PCA9548A I2C Mux (TSSOP-24)	1	\$1.72	\$1.72	Digi-Key PCA9548APWR
VL53L0X Time-of-Flight Sensor	6	\$14.95	\$89.70	Adafruit VL53L0X (3317)
AS7341 Color/Spectral Sensor	1	\$18.95	\$18.95	Adafruit breakout 4698
BNO055 IMU (gyro/accelerometer)	1	\$34.95	\$34.95	Adafruit breakout 2472
DC Brushed Gearmotor (37D, 6.3:1, 12V)	4	\$38.95	\$155.80	Pololu item #4747
2S Li-ion Battery Pack (~4000 mAh)	1	~\$60.00	~\$60.00	~7.4V 4250 mAh pack on BatterySpace \$63.19
XL4016 Buck Converter (8–10A)	1	~\$3.50	~\$3.50	Generic XL4016 module ₹251 (~\$3.4)
Push button switch, momentary	2	~\$0.50	~\$1.00	E.g. 6 mm tactile button (~\$0.3–0.5 each)
WS2812B NeoPixel LED	2	~\$1.00	~\$2.00	Addressable RGB LED (~\$1 each)
Subtotal (est.)	–	–	\$381.19	–
OpenMV Cam H7 R2 (vision, optional)	1	\$80.00	\$80.00	Camera for vision (planned)